

Relay

Turn what happens in an app into notifications people actually want

Core: first ~3 hours · Optional: next few hours · Fully local, no external services · Any stack

In one paragraph

Apps constantly tell people about things that concern them: *someone commented on your post, you were mentioned, you have a new follower*. In this exercise you will build two small pieces. **Chirp** is a pretend social app where made-up people do things.

Relay is a notification service that listens to what happens in Chirp, decides who should hear about it and what to tell them, respects each person's choices, and puts the notification in their in-app inbox.

How to read this brief

This is a product description, not a technical spec. It explains what to build, who it is for and how it should behave, but not how to build it; those decisions are yours. Everything runs on your own computer. There is no email, no text messaging and no outside service of any kind. The work is in two parts:

Part	Time	What it is
A. Core	First ~3 hours	Four features that together make a small but complete, working product.
B. Optional	Next few hours	Features that make Relay smarter and more pleasant to use. Pick the ones that interest you, in any order.

Relay is inspired by real notification platforms such as Courier, but you do not need to know any of them. Everything you need is in this brief.

1. Why a product like this exists

A **notification** is a short message an app shows a person because something happened that concerns them. Almost every app has them, and they usually start out simple: wherever something happens in the code, a developer adds a line that creates a notification. Over time, this becomes painful:

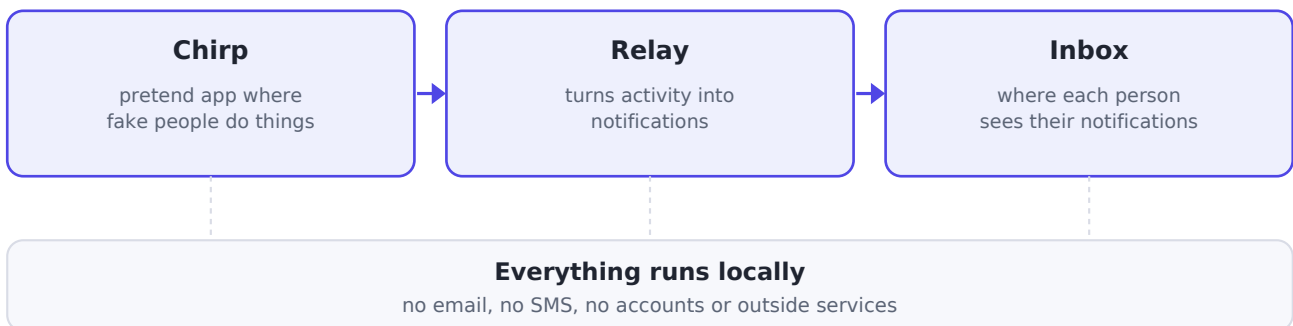
- **Changing the wording needs a developer.** The text lives inside the app's code, so fixing a typo means a code change.
- **Every feature does it differently.** Comments, likes and follows each create notifications in their own way, so they look and behave inconsistently.
- **People get notified about things they do not care about.** Someone turns off "like" notifications, but one part of the app forgets to check.
- **Nobody can say what happened.** When a user asks "why didn't I get notified?", no one can quickly check.

The idea behind Relay

Take notifications out of the app and put them in one place. The app only reports **what happened**, like a news wire. Relay is the newsroom: it decides who needs to know, writes the message, checks whether each person wants this kind of news, delivers it to their inbox, and keeps a record.

2. What you will build

Two separate pieces that talk to each other, plus the inbox where results appear:



Chirp, the pretend app

Chirp exists only to create realistic activity, so it should be as simple as possible. It has a few made-up people (for example Asha, Rahul, Meera and Dev) who can write posts, comment on posts, like posts, mention each other with an @name, and follow each other. Whoever is using Chirp picks which person they are "acting as", then does things as that person. Chirp does not need logins, real accounts or good looks.

The one rule that matters

Chirp never creates notifications itself. It only tells Relay what happened, for example "*Rahul commented on Asha's post*". Everything about notifications lives in Relay. If Chirp were replaced with a completely different app tomorrow, Relay should not need to change.

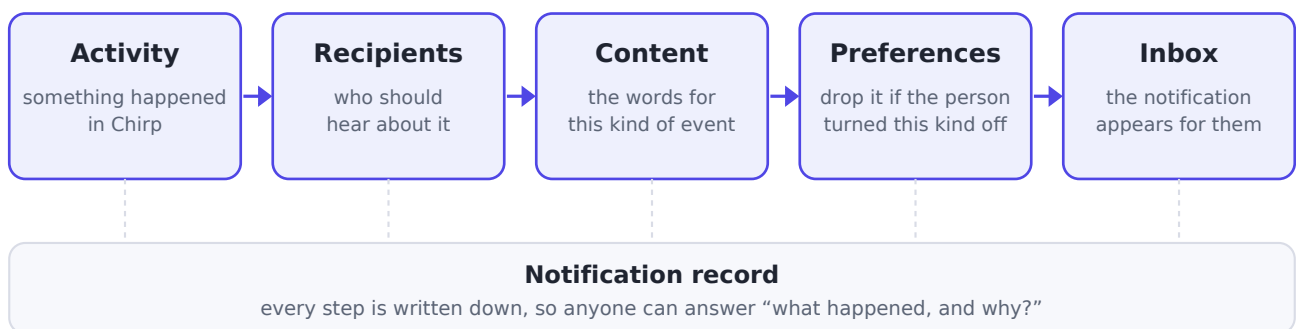
3. Who uses Relay

Three kinds of people care about Relay. Keep them in mind whenever the brief leaves a decision to you.

Who	Their role	What they want
The app developer	Builds an app like Chirp.	To report activity to Relay in one simple step and never write notification logic themselves.
The content owner	Decides what notifications say (often a product manager).	To change the wording of each kind of notification without asking a developer.
The recipient	A person using the app (Asha, Rahul...).	To be told about things that matter to them, not to be spammed, and to catch up in their inbox.

4. The key ideas

Relay is built from a handful of simple ideas. Every piece of activity moves through them from left to right:



Idea	What it means
Activity (event)	A fact about something that happened in the app, e.g. <i>Rahul commented on Asha’s post</i> . Each activity has a type (comment, like, mention, follow), who did it, and a few details such as the post or the comment text.
Recipient	A person who should be told. Usually this is obvious: the post’s author hears about comments on it, a mentioned person hears about the mention. People are never notified about their own actions.
Notification type	The kind of notification, matching the kind of activity: new comment, new like, mention, new follower. People make choices per type.
Content	The words for each notification type, with blanks filled in from the activity, e.g. “ <i>{actor} commented on your post: ‘{comment}’</i> ”. The content owner can change these at any time.
Preferences	The notification types each person has switched off. If Asha turned off likes, she never gets like notifications.
Inbox	Each person’s list of notifications, newest first, like the bell icon in most apps, showing which are new (unread).

Idea	What it means
Notification record	Relay's receipt for each activity: who was considered, who was notified, and who was skipped and why (for example, "Asha has turned off likes").

5. An example, start to finish

Asha has written a post on Chirp. She likes hearing about comments and mentions, but has turned off notifications for likes.

Step	What happens
1 Rahul comments	In Chirp, someone acting as Rahul comments "Great photo!" on Asha's post. Chirp tells Relay: <i>Rahul commented on Asha's post, comment "Great photo!"</i> . That is all Chirp does.
2 Relay works out who	The post belongs to Asha, so Asha should hear about it. Rahul wrote the comment, so he does not.
3 The words are written	Relay uses the content for new comments: "Rahul commented on your post: 'Great photo!'"
4 Asha's choices apply	Asha accepts comment notifications, so this one goes ahead.
5 It appears	Asha's inbox shows one new, unread notification.
6 Meera likes the post	Chirp reports it the same way. Asha has turned off likes, so she is skipped and nothing appears. The record notes why.
7 Asha catches up	Switching to Asha in Chirp, she opens her inbox, reads the comment notification, and it is marked as read.

When Part A is done, someone following your instructions should be able to recreate this exact story.

Part A — Core features (first ~3 hours)

These four features together make the smallest version that is genuinely useful. Aim to finish all four before starting anything optional. Chirp and the inbox need to be usable in a web browser; anything else (how content is edited, how records are viewed) can be as simple as you like.

A1. Chirp, the pretend app

As someone trying out the system, I can act as different people and do things in Chirp, so that there is real activity for Relay to respond to.

What must be true

- Chirp has a few made-up people and lets me choose who I am acting as.
- I can write a post, comment on a post, like a post and mention someone. (Following is optional.)
- Every action is reported to Relay as “what happened”. Chirp never creates notifications itself.
- Chirp keeps working even if Relay is not running; activity simply goes unnotified.

A2. Turn activity into notifications

As the app developer, I only report what happened, and Relay decides who to tell and what to say.

What must be true

- Relay works out the recipients for each kind of activity, and never notifies people about their own actions.
- Each notification type has its own wording, with blanks filled in from the activity.
- The content owner can change the wording, and new notifications use it, without any change to Chirp.
- Activity Relay does not recognise is rejected with a clear explanation rather than silently ignored.

A3. Respect people's choices

As a recipient, I can choose which kinds of notifications I get.

What must be true

- Each person can switch any notification type off and back on.
- A switched-off type is never delivered to that person; the record shows they were skipped and why.
- New people start with sensible defaults. Write down what you chose and why.

A4. Inbox and records

As a recipient, I can see my notifications. As anyone at the company, I can check why someone was or was not notified.

What must be true

- Each person has an inbox showing their notifications, newest first, with unread ones clearly marked and an unread count.
- Opening a notification, or marking it read, updates the count. There is also a way to mark all as read.
- A person only ever sees their own inbox.
- For any activity, one can see who was notified and who was skipped, with the reason, in plain language.

Part B — Optional features (next few hours)

Each of these makes the product more like a real one. They are independent, so choose the ones you find most interesting. Doing two well is better than doing five partly.

B1. Autopilot

Clicking around by hand only creates a little activity. Real apps are busy all the time.

What must be true

- Chirp can be switched to autopilot, where the made-up people do random things on their own every few seconds.
- It can be paused, and its speed can be changed.
- Relay keeps up: nothing is lost, even when activity is fast.

B2. Live inbox

Today a person must refresh the page to see new notifications.

What must be true

- New notifications and the unread count appear on their own, a moment after the activity happens.

B3. Grouping

If ten people like Asha's post in a few minutes, ten separate notifications is noise.

What must be true

- Similar notifications about the same thing are combined, e.g. "Rahul and 4 others liked your post".
- The grouped notification updates as more activity arrives, and becomes unread again when it does.

B4. Followers

Following someone means wanting to know when they post.

What must be true

- People can follow and unfollow each other in Chirp, and being followed creates a notification.
- When someone writes a post, each of their followers is notified.
- Decide what should happen if someone has many followers, and note it in your README.

B5. Mute and snooze

Sometimes people want a break without changing their settings for good.

What must be true

- A person can mute notifications about a specific post (for example a busy thread).
- A person can snooze all notifications for a while. Decide what happens to activity during the snooze, and explain why.

B6. No duplicates

If Chirp has a hiccup and reports the same activity twice, Asha should still be notified only once.

What must be true

- Relay recognises a repeated report of the same activity and does not notify anyone again.

What to send us

- A code repository whose history shows how the work progressed.
- A way to start everything on a fresh machine with a single command, with no accounts or paid services needed.
- A few made-up people in Chirp and wording for each notification type, ready to use from the start.
- A short README: how to run it, how to recreate the example in section 5, which optional features you chose, any assumptions you made, and what you would do next with more time.
- If you used AI tools, a sentence on where.

If anything in this brief is unclear, make a sensible decision and write it down in your README.